

Mimanshi Sharma

+91 93153-00494 | sharma.mimanshi24@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

AI/ML Engineer building LLM applications, RAG pipelines, and multi-agent systems in Python, FastAPI, and GCP.

EDUCATION

SRM Institute of Science and Technology

B.Tech in Computer Science and Engineering – AI/ML Specialization

• CGPA: 8.1 / 10

Kattankulathur, Chennai

June 2022 – Present

EXPERIENCE

DRDO — DYSL-AI Lab

AI Research Intern

Bengaluru, Karnataka

January 2026 – Present

- Built an automated Python **data pipeline** (yt-dlp, ffmpeg, ytmusicapi) to curate a 600-song audio dataset, achieving a **48× per-row speedup** and saving 33 hours of manual effort.
- Designed a **data validation and quality assurance** system **detecting 19.7% broken records** and **auto-correcting 60%**, reducing manual review from 100% to 7.9% of the dataset.
- Benchmarked** three audio LLMs (**SALMONN-7B**, **Audio Flamingo Next**, **Qwen2-Audio**) on the dataset for gender classification, conducting comparative **model evaluation** and **failure analysis**.
- Stack:** Python · PyTorch · Hugging Face Transformers · Data Pipelines · Data Validation · Audio Processing · Model Evaluation & Benchmarking · Multi-GPU Inference (Quadro RTX 6000/8000)

PROJECTS

Memory AI – Persistent RAG Assistant — [Live Demo](#)

- Built a production-ready **RAG** chatbot with LLM-powered (**Gemini API**) persistent memory using **FastAPI**, React, Firestore, and **vector embeddings** for **semantic search**-based context retrieval.
- Engineered **context window management** for multi-turn dialogues via cosine similarity search over a **vector database** of embedded conversation history; deployed full-stack (React 18 + FastAPI) to production.

LLM Multi-Agent Pipeline for Bias Detection — [GitHub](#)

- Built a **multi-agent AI system** (4-agent **orchestration pipeline**: Generator → Auditor → Refiner → Verifier) to detect and fix biased language across 4 demographic categories using **Ollama** and **agentic workflows**.
- Designed modular **AI agent orchestration** with **prompt engineering** and automated **evaluation workflows**, identifying **hallucination** failure modes in agent-to-agent communication.

Coral Reef Classification using ResNeSt+CSA

- Built a deep learning model for automated coral health classification from underwater imagery using **ResNeSt-50** with a novel **Cross-Spatial Attention (CSA)** module
- Achieved **92.4% accuracy** (92.0% precision, 91.7% recall) on 7,384 images; outperformed ResNet-50 (87.2%) and ResNeXt-50 (89.1%); ablation study validated CSA contribution
- Deployed **Flask** web application for real-time inference; paper **accepted at ICCBI 2026 International Conference**

CERTIFICATIONS

Google Cloud/DeepMind: ADK Agents (Intermediate), SLM Training (Advanced), Intro to Generative AI | AWS Academy: ML Foundations, Cloud Operations

TECHNICAL SKILLS

Generative AI: LLMs, RAG, AI Agents, Multi-Agent Systems, Agentic Workflows, Prompt Engineering, Context Retrieval, Semantic Search, Vector Embeddings, Model Evaluation & Benchmarking

Frameworks & Libraries: Hugging Face Transformers, PyTorch, FastAPI, Ollama, Gemini API, OpenAI API, Flask

ML/Data: Data Pipelines, Data Validation & Quality Assurance, Deep Learning, Computer Vision, Audio Processing, Failure Analysis

Cloud & DevOps: GCP, Cloud Run, Docker, Firestore, Git, Linux

Languages: Python, Bash